London Metropolitan University

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## Module Code & Module Title

**CU6051NI Artificial Intelligence**

**25% Individual Coursework**

**Submission: Proposal**

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**Submitted To: Er. Roshan Shrestha**

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|---|---|
| **GitHub Link** | [https://github.com/SabinAdhikarii/Credit-Card-Default-Detection-.git](https://github.com/SabinAdhikarii/Credit-Card-Default-Detection-.git) |

*I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.*

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# 1. Introduction

## 1.1 Introduction to the Topic

Credit card default prediction is a vital area in monetary and financial risk management. The increase in digitalization of have directly caused in growth of usage of credit card usage worldwide, financial institutions are facing increasing challenges in identifying customers who may fail to repay their obligations. Nepal not being very far, have also started digitalization in cards distributing credit card for first time in 1990 (the HRM, 2023). Since the day, Nepal also has distributed almost more than 318,428 credit cards in Nepal (INVESTOPAPER, 2025). The credit cards have been in an easy access to the people nowadays but its actually very hard to interpret if the user will be able to pay back the used credit amount or not.

For the solution of this problem, Artificial Intelligence (AI) can be used. This coursework allows us to make an AI system that can predict if the user can actually pay back the credit amount or not. For this a model will be trained using the past records of the user with various factors and predict defaults, enabling proactive risk mitigation.

## 1.2 AI Concepts Used In The Coursework

The project leverages several core AI concepts, the model will be first loaded and trained on labeled data as supervised learning methods. The classification of the data will be done using the classification algorithms like Logistic Regression, Random Forest, Gradient Boosting or others. The concept of feature engineering will be used for transforming raw demographic and financial data into meaningful inputs. For the cases like imbalanced or inconsistent data, data cleaning and handling techniques will be used and at last the model will be evaluated on using evaluation metrics if the prediction is satisfying the system is deployed otherwise the parameters are changed and evaluated again.

## 1.3 Current Scenario and Issues/Challenges

The credit card default can be refereed as using or taking the benefits from the credit card using it more the purposes like shopping, paying, and getting exclusive offers but not paying at time or never paying back to the back. This problem exists both in global scale and at local levels.

Table 1: Comparison of Global Challenges & Local Challenges

| Global Challenges | Local Challenges (Nepal Context) |
|---|---|
| Increasing in fraudulent activities and defaults leading to billions in losses annually. | Increasing in debt of a person due to lack of awareness about repayment obligation. |
| Increasing reliance on credit cards for online transactions. | Increasing reliance on credit card for instant payment for goods or services. |
| Regulatory pressure on banks to maintain financial stability. | Banks face challenges in assessing creditworthiness due to incomplete financial histories. |
| Fraudulent activities and defaults leading to billions in losses annually. | Fraudulent activities as person unable to repay the amount to bank in time. |

## 1.4 Problem Statement

Banks or any other financial institution that provide credit card and allows user to use credit up to some extents are facing huge losses and backlashes due to poor identification of the person who is likely to pay or not pay the credit amount. Traditional credit scoring system often rely on limited demographic or historical repayment data, which fails to capture the complex behavior and financial pattern that could influence default risk.

The imbalanced and inconsistent records where the number of non-default cases far outweighs default cases are also the challenge of this coursework. These imbalances made it harder for conventional models to detect rare but critical default events, leading to poor recall and precision of the model.

## 1.5 Project as a Solution

The proposed project is “Credit Card Default Detection System”, for solving the chances of frauds by early detection of the customer who is likely to be at high risk. The system will have a model that will be trained on a transaction data of the bank specifically of the credit card payment history, billing data, nature and behavior of the client. The model will analyze and dataset, finding the clients with high chance of risk. The system will allow back to identify the risky clients early and take any necessary actions, lower the credit limits, or add a friendly reminder for payments.

The AI system will leverage the banks efficiency reducing both risk and loss amounts and successfully gaining more profits gradually.

## 1.6 Aims and Objectives

**Aims**

The aim of the project is to develop a supervised learning solution for predicting credit card defaults using AI model.

**Objectives:**

- To collect and preprocess financial datasets.
- To train the AI models and test them.
- To evaluate performance using precision, recall, and F1-score, etc.
- To provide conceptual diagrams and pseudocode for implementation.

## 1.7 Dataset Selection for the Project

**Name**: Default of Credit Card Clients (Taiwan)

**Source**: UCI Machine Learning Repository

**Size**: 30,000 rows

**Features**: 24 variables

# 2. Background

## 2.1 Project Overview and Justification

The project is titled as **“Credit Card Default Detection”.** This project is an essential project required in the domain of finance to detect the default of credit so that the clients who is the red zone or in the verge where they might not be able to repay the obligations can be identified and necessary actions like freezing the current credit status, limiting the credit amount, stopping the payments or more can be done. The project focusses on implementing supervised learning as the dataset, itself provides labeled outcome as default or non-default. This approach is the standard industry align practices which even offer measurable accuracy improvement. The system tends to implement at least of three algorithms. They are: Logistic Regression, Decision Tree and Random Forest. The system will evaluate the recall, precision and F1 score of all the models and evaluate the model with the highest accuracy. The performance of any classification algorithm can be influenced by the dataset. Hence, all three algorithm’s output will be analyzed comparatively.

The justification for this project lies in its real-world applicability:

- It helps banks and other financial institution to reduce their debts and losses by identifying high-risk clients early.
- It supports regulators in maintaining financial stability.
- It benefits customers by ensuring fairer lending policies and proactive support.

### 2.2 Background / Literature Review

Credit card default prediction is a serious problem in the financial domain. Banks and financial institutions face significant risk when client fails to repay their credit amount. Before the development of our project, it is important to do research if the project or complete or partially have already been done by other people also. During my research to check if any other similar project is already available or not. I found some of the similar projects. The project is done on similar basis but not the exact project had done. The ground for the development of the project were same for default credit card detection but they have used different dataset, different algorithms and techniques. I have shortlisted three similar projects. They are:

### 2.2.1 Consumer Credit-Risk Models via Machine-Learning Algorithms

**"Consumer Credit Risk Models via Machine Learning Algorithms"** is the one of the similar projects done by Amir E. Khandani, Adlar J. Kim, and Andrew W. Lo. The project mainly focuses on implementing machine learning to big financial data for customer credit risk. The tree-based algorithm (CART-like) is used to build models and compared with traditional methods (HAL-SH, 2010). The authors compared the performance of these machine learning models against credit scoring method and demonstrated the machine learning superior predictive accuracy and robustness. This project by Khandani, Kin, and Lo(2010) is similar to our project as both project uses machine learning models to find the accuracy of the credit default. Also, both projects are emphasizing the importance of behavioral and finance features in building robust models.

### 2.2.2 Random Forest for peer-to-peer lending defaults.

Another important project in the field of credit default detection which is also similar to our project was conducted by Malekipirbazari and Aksakalli in 2015 titled as 1. Random Forest for peer-to-peer lending defaults. This study was more aligned toward growing peer-to-peer lending platforms, where individual lend money directly to the other individuals without traditional banking intermediaries (Y. N. Ifriza, 2015). The author recognized default risk in peer-to-peer and applied Random Forest algorithm to ensemble it with decision tree to improve the accuracy and reduces the overfitting. This project concludes that Random Forest is a great choice for the project like credit risk detection (Aksakalli, 2018).

### 2.2.3 Credit Risk Prediction Using Machine Learning and Deep Learning

This project serves the similar project operations as it compares the several algorithms like Logistic regression, AdaBoosr, XGBoost, LightGBM, and Neural Networks so that the model could predict the best accurate ouput (Chang, 2024). This research was done by Victor Chang, Sharuga Sivakulasingam, Hai Wang, Siu Tung Wong, Meghana Ashok

Ganatra, and Jiabin Luo in 2024. The project is similar to ours as it focuses on credit card default prediction, uses supervised learning and perform ensemble methods.

### Review and Comparison of Similar Projects

Table 2: Comparison Between Similar Project

| Project | Algorithm | Dataset | Accuracy | Strengths | Weaknesses |
|---|---|---|---|---|---|
| Khandani et al. (2010) | ML models (CART-like trees) | US consumer credit data | ~80% | Rich dataset, captures behavioral patterns | Complex preprocessing, computationally heavy |
| Malekipirbazari & Aksakalli (2015) | Random Forest | Peer-to-peer lending platform | ~85% | High accuracy, robust to imbalance/noise | Interpretation is low. |
| Victor Chang et al. (2024) | ML & Deep Learning (Neural Networks) | Credit card customer dataset | ~85–90% | Ensemble + deep learning superior recall and ROC-AUC | Requires high computational resources, lower interpretability |

# 3. Solution

## 3.1 Explanation of the Proposed Solution

The proposed solution to the problem of credit card default prediction is a development of a machine learning model that could analyze the customer or client's demographic data, repayment history and billing records to predict if a client is likely to default or mot. Here in this system supervised learning will be used meaning the model will learn from the labeled data sets to train to identify the underlying patterns and relationship (Ivan Belcic, 2025). This solution helps banks and financial institutions to proactively identify high-risk clients, reduce financial losses and improve credit risk management system smoothly.

## 3.2 Explanation of AI algorithms Used

The project specifically focused on implementing three classification algorithms for the evaluation of the AI model developed. After allocating the training and testing data from in the ration 80:20, the algorithms are used to learn from the training and testing data, it usually recognizes patterns and make prediction (GeeksforGeeks, 2023). Since, our project majorly falls under identifying the likelihood of a person to check if he/she could actually repay the obligation or not, the classification technique is best. Under the classification technique, Logistic Regression, Random Forest and Gradient Boosting are the three algorithms being used in our project.

### 3.2.1 Logistic Regression

**Logistic Regression:** Logistic regression, though having regression in its name falls under classification. It is an algorithm under supervised machine learning used for classification problem. It is used for binary classification problems like True/False, Yes/No, in our case defaulter/non-defaulter. It works by concerting the inputs into probability values between 0 and 1 (GeeksforGeeks, 2025).

In our project, it serves as a baseline model that interprets and provides insights into default risk.

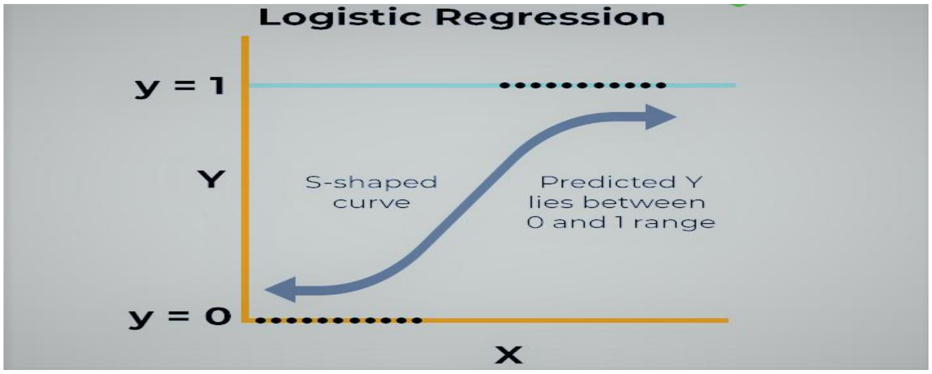



Figure 1: Logistic Regression

### 3.2.2 Random Forest

**Random Forest:** Random Forest is an AI algorithm that uses ensemble techniques to make the prediction from decision tree. As the name suggests, it is the combination of large number of trees. It creates a large number of decision-tree trained on different dataset and combined all of them while making prediction (tutorialspoint, 2023).

In our project, this algorithm will be used for making prediction based on the decision made by multiple decision tree. The final prediction will be made combining the prediction of all decision tree.

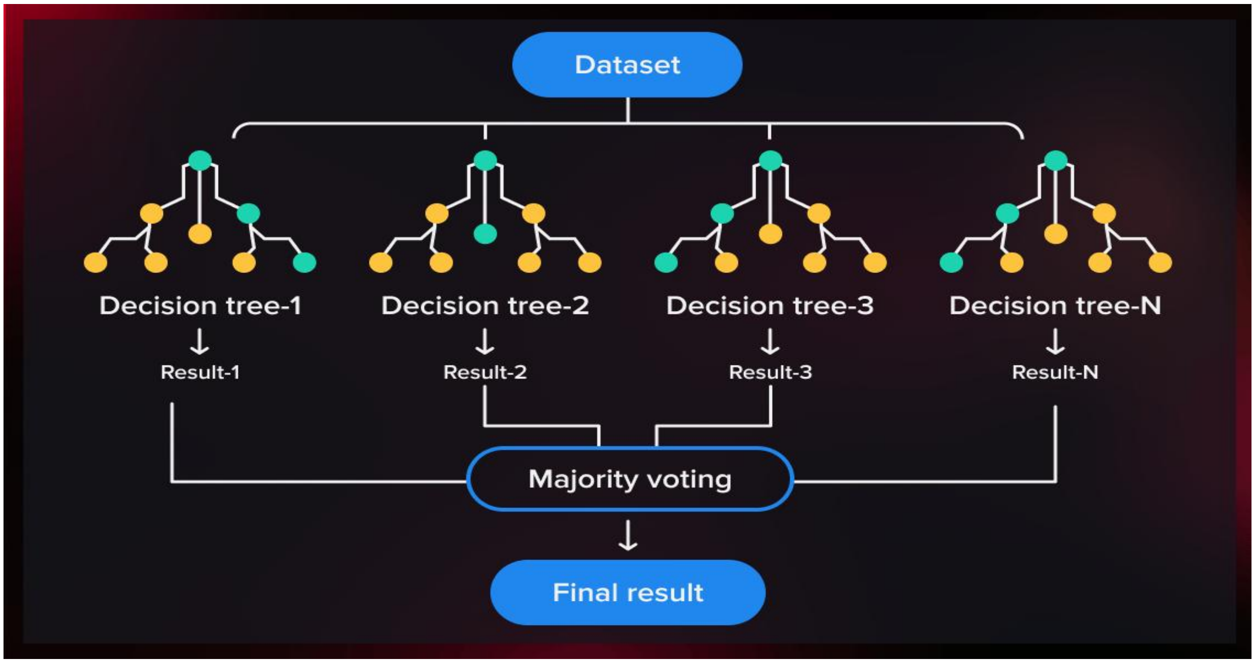



Figure 2: Random Forest

### 3.2.3 Gradient Boosting

**Gradient Boosting:** Gradient Boosting, a supervised machine learning algorithm which produces accurate and precise prediction by combining multiple decision trees into a single model (Bryah Clark, 2024).  The algorithm work by combining the weak prediction into single ensemble. This algorithm minimizes the loss function and improves the predictive accuracy (Bryah Clark, 2024)

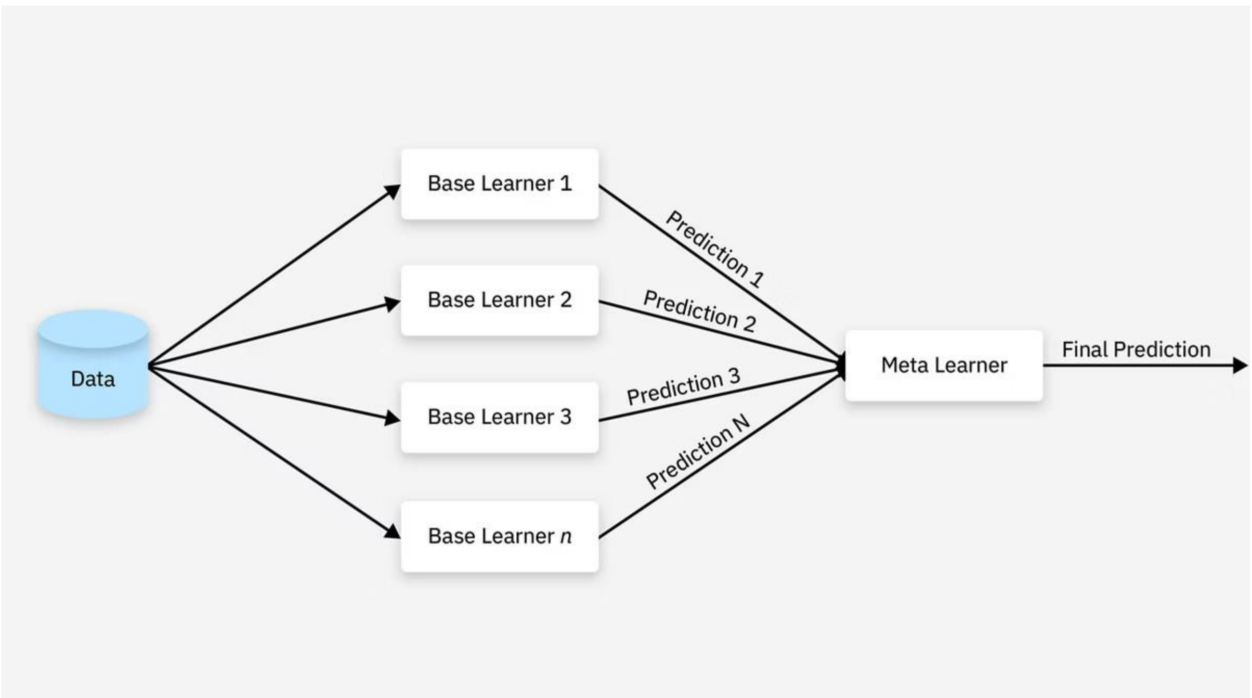
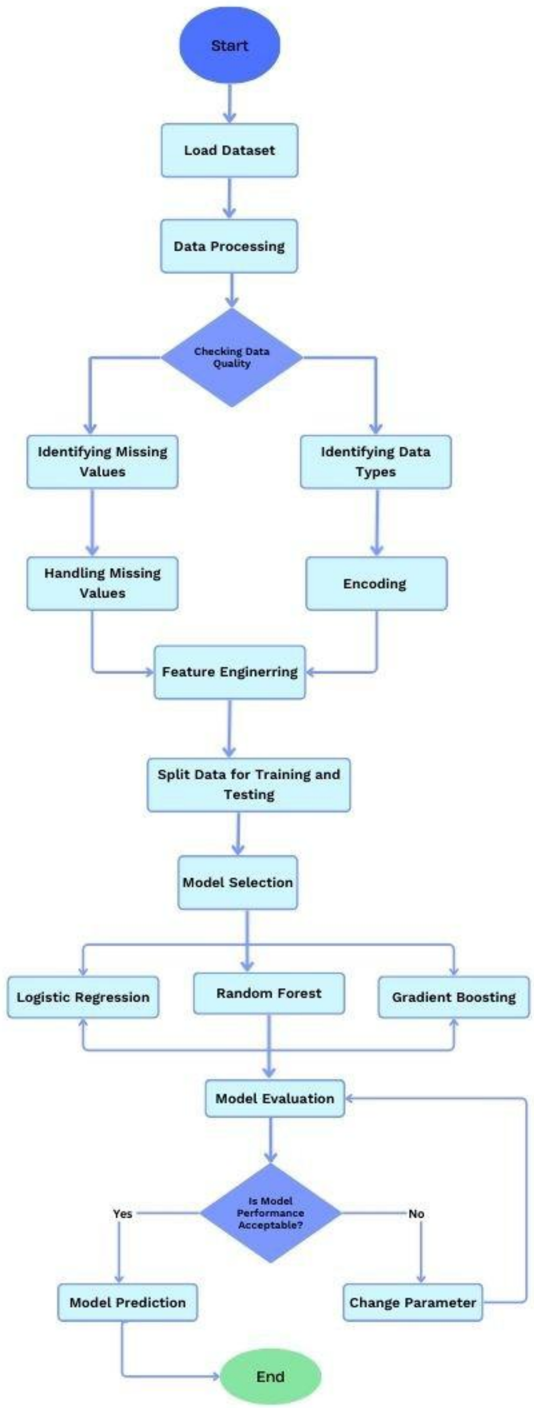



Figure 3: Gradient-Boosting

## 3.3 Diagrammatical Representation

### 3.3.1 System Flowchart

Figure 4: System Flowchart



### 3.3.2 Workflow Diagram

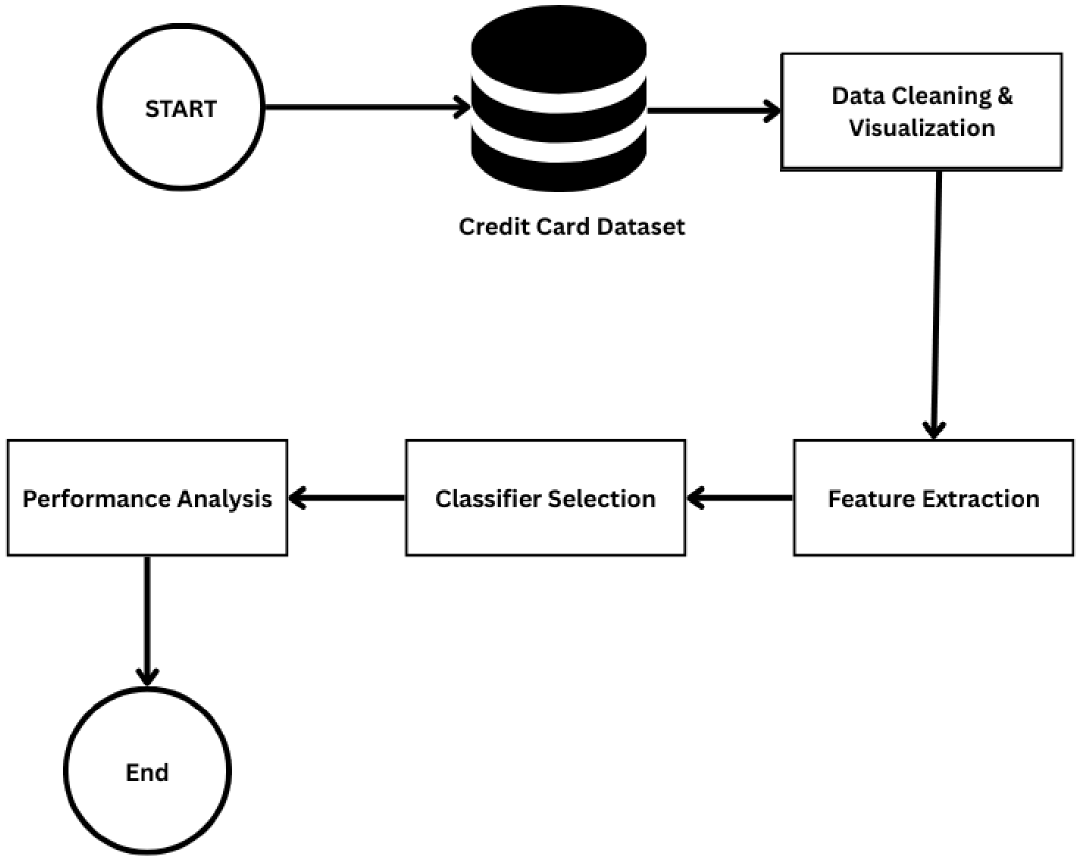



Figure 5: System Workflow

### 3.4 Pseudocode

**BEGIN**

Load the Credit Card Default dataset

Process the data:

  Handle missing values

  Identifying data type

  Encode string values into numeric

  Selection of required features

Splitting Dataset into train-test sets

For all chosen algorithm do:

  Train the model on the training data

    Logistic Regression

    Random Forest

    Gradient Boosting

  Test the trained model on the testing data

Model Evaluation using:

    Accuracy

    Precision

    Recall

    F1-score

    ROC-AUC

    Increase the model accuracy by changing the parameter

Select the best-performing model

Predict default status for new clients

Display results

**End**

## 4. Conclusion

The project serves as solution to the problem of credit default using the concept of machine learning and artificial intelligence algorithm. The coursework includes the development of a system with at least of 3 algorithms to compare their analysis. During the development of the project, the concepts like supervised learning, models and other required things are studied and implemented properly. The research papers journals and articles have also studied that helps us understand the work done in this area.

Overall, the system is eligible for predicting the best outcome as it compares the prediction f other models. The concept of ensembling the predicted data is also learned during the coursework. The methodology followed a structure ML pipeline, including dataset loading, processing, training, evaluation ensuring reliable results of the dataset. The project solves the problem of financial institution and banks by reducing financial losses and identifying the risky clients causing the debt to minimize.

### 4.1 Future Works

The future works for the system will most be integrating it with the banking system or financial companies to work in real world.  Expanding more features and exploring neural networks for high performance, accepting multiple requests at a time.

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